

Viscous Obstructions to Prescribed Sparse Source Chains in Navier–Stokes Flow

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Abstract

We study two prescribed sparse Fourier source chains along one real, cyclically symmetric, incompressible, unforced Navier–Stokes solution on a fixed three-dimensional torus with viscosity one. The contribution of a selected input projection is defined by its full Duhamel integral; its remainder includes the existing target and every omitted interaction. For an adjacent-sum chain, the frequency differences remain fixed and the selected source is bounded by a constant times the inverse square of the target frequency. A pairwise source-dominance condition then forces the newly received layer’s gradient to decay. For a cross-edge chain, the frequency differences grow, but two heat integrations yield an explicit geometric gradient decay under the two stated source-dominance conditions, with the first holding throughout the second interaction interval. At a common time, conditions at every layer imply absolutely summable first-derivative tails for the corresponding chain projection. We also give a general short-time source-lag lemma and a fixed algebraic-data example in which a difference output precedes the selected receiver’s self-supply. The source conditions are hypotheses to be checked on the actual solution. These results concern specified projections and do not establish regularity or singularity of the full Navier–Stokes flow.

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1 Introduction

A frequency produced by a quadratic interaction need not subsequently become the dominant source of another frequency. This distinction matters when a finite interaction pattern is proposed as one step of a repeated construction. Frequency arithmetic describes possible outputs; source attribution must also account for the evolution of the input, viscous damping, and all other contributions to the target.

We examine two fixed recurrences beginning with the cyclic orbit of $(3, 4, 6)$. Both are tested on projections of one complete Navier–Stokes solution. No projected field is assumed to solve a closed finite-dimensional system. The adjacent-sum recurrence doubles an axial component while preserving every within-layer frequency difference. In this case incompressibility replaces the derivative frequency in the selected interaction by a fixed difference. The cross-edge recurrence expands those differences, but both of its prescribed intermediate frequency types remain comparable to the sending frequency. Two full heat integrations then give a different decay estimate.

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The main results are Theorems 3.1 and 4.2. Energy and heat estimates give unconditional bounds for the adjacent-sum source and the first cross-edge source. The second cross-edge source bound additionally requires the first source condition throughout its integration interval. Bounds for the actual targets use precise pairwise comparisons between the complete remainder and the designated source. These conditions form a sufficient attribution test, not a necessary definition of every possible transfer mechanism. We distinguish reception at different times from conditions holding for all layers at one time. Only the latter permits a simultaneous chain estimate.

Section 5 treats a complementary finite-time obstruction. A target with a nonzero first-order response cannot, on a sufficiently short original time interval, be predominantly supplied by the self-interaction of a projection that is third order in time. We prove this as a general source-lag statement, including every truncated integration interval, and apply it to the fixed algebraic initial datum of the companion manuscript [3]. This example links finite source selection to the need for a separate test of subsequent supply.

The energy identity, Fourier representation and heat estimates used here are standard; see, for example, [1]. Fourier triad interactions and their channel-dependent cancellations are classical topics; see Waleffe's helical analysis [2]. Here the norm comparison concerns velocity coefficients, and is not a claim of positive net energy flux. The purpose is to give complete estimates for the two stated recurrences, with explicit constants and time quantifiers. In particular, the statements do not refute a construction for a different fluid equation, and they do not exclude other input sets, growing numbers of modes, or mechanisms outside these source conditions.

2 The full equation and source attribution

2.1 Fixed setting and conventions

Throughout the two chain theorems,

$$\Omega = \mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3, \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0. \quad (2.1)$$

The viscosity is exactly one. Let u, p be a real classical solution on $[0, T)$, smooth on every compact time subinterval, with one fixed smooth, zero-mean, divergence-free initial datum $a = u(0)$. No conclusion about the size of T is assumed. Define

$$u(x, t) = \sum_{k \in \mathbb{Z}^3} u_k(t) e^{ik \cdot x}, \quad \|f\|_2^2 = (2\pi)^{-3} \int_{\Omega} |f|^2 = \sum_k |f_k|^2, \quad M = \|a\|_2. \quad (2.2)$$

Complex coefficient lengths are Hermitian Euclidean lengths; the dot products in the convolution are complex bilinear. Reality and incompressibility give $u_{-k} = \overline{u_k}$ and $k \cdot u_k = 0$. For $k \neq 0$, put $\mathbb{P}_k = I - k \otimes k / |k|^2$ and

$$B(f, g)_k = -i\mathbb{P}_k \sum_{p+q=k} (f_p \cdot q) g_q, \quad B(f, g)_0 = 0. \quad (2.3)$$

The projected equation is $u_t - \Delta u = B(u, u)$. Thus every source below retains the complete Leray projection; a vanishing pressure term in the energy identity does not remove pressure from this equation.

For a symmetric frequency set $F = -F$, let P_F be coefficient projection and write

$$\|f\|_{\mathcal{A}_0} = \sum_k |f_k|, \quad |f|_{\mathcal{A}_1} = \sum_k |k| |f_k|. \quad (2.4)$$

These expressions are finite on the finite projections used below. In particular,

$$\|\nabla f\|_{L^\infty} \leq |f|_{\dot{A}_1}, \quad \|B(f, g)\|_{\mathcal{A}_0} \leq \|f\|_{\mathcal{A}_0} |g|_{\dot{A}_1}. \quad (2.5)$$

The matrix norm in the first inequality is Frobenius. Both inequalities follow by taking absolute values of the complete Fourier sums and using $\|\mathbb{P}_k\|_{\text{op}} \leq 1$.

2.2 Cyclic symmetry and the energy bound

Let $R(x_1, x_2, x_3) = (x_3, x_1, x_2)$ and assume

$$a(Rx) = Ra(x). \quad (2.6)$$

Orthogonality gives $\mathbb{P}_{Rk} = R\mathbb{P}_k R^\top$. Applying R to space and velocity therefore preserves the full equation. Classical uniqueness on each compact existence interval implies

$$u(Rx, t) = Ru(x, t), \quad u_{Rk}(t) = Ru_k(t). \quad (2.7)$$

For completeness, uniqueness follows from the difference equation for two classical solutions: its L^2 energy is bounded by a constant times the L^∞ norm of the gradient of one solution times the difference energy. That coefficient is bounded on each compact interval, so Gronwall applies. This also justifies applying uniqueness to the rotated solution.

Multiplication of (2.1) by u , followed by periodic integration, gives

$$\|u(t)\|_2^2 + 2 \int_0^t \|\nabla u(r)\|_2^2 dr = M^2. \quad (2.8)$$

The convection integral is an integral of a divergence, and the pressure integral vanishes by incompressibility. Consequently, if $D = \{\pm k, \pm Rk, \pm R^2k\}$ consists of six distinct frequencies and $h = P_D u$, then

$$6|u_k(r)|^2 = \|h(r)\|_2^2 \leq M^2. \quad (2.9)$$

This bound holds for the actual complex coefficients, with no prescribed phase, polarization line, or nonvanishing assumption.

2.3 Source integrals, remainders and pairwise conditions

Let $h = P_D u$ be any actual finite projection and let $F = -F$ be a finite target set. For $0 \leq s < t < T$, define

$$S_F^h(t; s) = \int_s^t e^{(t-r)\Delta} P_F B(h(r), h(r)) dr, \quad Z_F^h(t; s) = P_F u(t) - S_F^h(t; s). \quad (2.10)$$

Writing $z = u - h$, the full Duhamel formula gives the exact identity

$$\begin{aligned} Z_F^h(t; s) &= e^{(t-s)\Delta} P_F u(s) \\ &\quad + \int_s^t e^{(t-r)\Delta} P_F [B(h, z) + B(z, h) + B(z, z)](r) dr. \end{aligned} \quad (2.11)$$

The first term contains the target already present at s . The other terms contain all frequencies outside D , both ordered cross interactions, and their self-interactions. Formula (2.11) is not a closure assumption for h .

Definition 2.1 (Pairwise source dominance). For a nonzero k with $\{k, -k\} \subset F$, the source condition at (s, t) is

$$\left\| P_{\{\pm k\}} Z_F^h(t; s) \right\|_{\mathcal{A}_0} \leq \frac{1}{4} \left\| P_{\{\pm k\}} S_F^h(t; s) \right\|_{\mathcal{A}_0}. \quad (2.12)$$

Conditions for a layer always mean (2.12) for every one of its positive–negative frequency pairs, with the same starting time within that layer.

Both fields are real, so a pair norm equals twice its representative coefficient length. The triangle inequality therefore yields

$$|u_k(t)| \leq \frac{5}{4} |S_F^h(t; s)_k| \quad (2.13)$$

when the pair condition holds. No division by the source is needed. A zero source together with the condition forces a zero target; it does not certify effective nonzero reception. A condition on a whole-band norm cannot in general be substituted for the separate pairwise conditions.

3 The adjacent-sum chain

Indices j below are taken modulo three. Set

$$k_j^{(0)} = R^j(3, 4, 6), \quad k_j^{(n+1)} = k_j^{(n)} + k_{j+1}^{(n)}, \quad D_n = \{\pm k_j^{(n)} : 0 \leq j < 3\}, \quad h_n = P_{D_n} u. \quad (3.1)$$

Let $S_{n+1}(t; s) = S_{D_{n+1}}^{h_n}(t; s)$ and $Z_{n+1} = h_{n+1} - S_{n+1}$, as in (2.10).

Theorem 3.1 (Adjacent-sum source and target bounds). *Assume the full setting (2.1)–(2.6). Put*

$$L_n^2 = \frac{169 \cdot 4^n + 14}{3}. \quad (3.2)$$

For every $n \geq 0$, every existing interval $0 \leq s < t < T$, and every $k \in D_{n+1}$,

$$|S_{n+1}(t; s)_k| \leq \frac{\sqrt{14}}{3} M^2 \frac{1 - e^{-L_{n+1}^2(t-s)}}{L_{n+1}^2}. \quad (3.3)$$

If the three target pairs satisfy Definition 2.1 on this interval, then

$$\|\nabla h_{n+1}(t)\|_\infty \leq |h_{n+1}(t)|_{\mathcal{A}_1} \leq \frac{5\sqrt{14}}{2} \frac{M^2}{L_{n+1}}. \quad (3.4)$$

Fix one time $t < T$. If for every $n \geq 0$ there exists $s_n(t) \in [0, t)$ such that all three conditions for that layer hold on $[s_n(t), t]$, then for every integer $N \geq 0$,

$$\sum_{m \geq N} |h_m(t)|_{\mathcal{A}_1} \leq \frac{5\sqrt{42}}{26} M^2 2^{-N}. \quad (3.5)$$

The chain projection $v_{\text{sum}}(t) = \sum_{m \geq 0} h_m(t)$ and its first spatial derivatives converge absolutely and uniformly, and

$$\|\nabla v_{\text{sum}}(t)\|_\infty \leq \sqrt{366} M + \frac{5\sqrt{42}}{26} M^2. \quad (3.6)$$

Proof. Write $A = (13/3)(1, 1, 1)$ and $w_j = k_j^{(0)} - A$. Direct dot products give

$$\sum_j w_j = 0, \quad A \cdot w_j = 0, \quad |A|^2 = 169/3, \quad |w_j|^2 = 14/3, \quad w_i \cdot w_j = -7/3 \quad (i \neq j). \quad (3.7)$$

Since $w_j + w_{j+1} = -w_{j+2}$, induction in (3.1) gives

$$k_j^{(n)} = 2^n A + (-1)^n w_{j+2n}, \quad |k_j^{(n)}| = L_n, \quad |k_i^{(n)} - k_j^{(n)}| = \sqrt{14} \quad (i \neq j). \quad (3.8)$$

Integrality follows from the recurrence. The three transverse vectors have Gram matrix with diagonal $14/3$ and off-diagonal $-7/3$; its kernel is spanned by $(1, 1, 1)$. In any linear dependence among the three $k_j^{(n)}$, the transverse component thus makes the coefficients equal, and the nonzero axial component then makes them zero. The three positive vectors are independent. Their coordinate sums are $13 \cdot 2^n$; the negative vectors have the opposite sums. Hence every layer has six distinct modes, and distinct layers are disjoint.

Fix $m = k_j^{(n+1)} = p + q$, with $p = k_j^{(n)}$ and $q = k_{j+1}^{(n)}$. A mixed-sign input pair has coordinate sum zero and a negative pair has a negative sum, so neither can produce m . Independence of the positive vectors shows that the only positive pairs producing m are (p, q) and (q, p) , including exclusion of a repeated input. Consequently the entire selected Fourier source is

$$B(h_n, h_n)_m = -i\mathbb{P}_m[(U \cdot q)V + (V \cdot p)U], \quad U = u_p(r), \quad V = u_q(r). \quad (3.9)$$

Incompressibility gives $U \cdot q = U \cdot (q - p)$ and $V \cdot p = V \cdot (p - q)$. Using (2.9), the full two-term coefficient is bounded by

$$|B(h_n, h_n)_m| \leq 2\sqrt{14}|U||V| \leq \frac{\sqrt{14}}{3}M^2. \quad (3.10)$$

This estimate allows arbitrary actual complex phases and uses both ordered terms. The negative targets obey the same bound by conjugation.

The target heat multiplier is exactly $e^{-L_{n+1}^2(t-r)}$. Integrating (3.10) along the actual history gives (3.3). When the pair condition holds, (2.13) gives

$$|u_k(t)| \leq \frac{5\sqrt{14}}{12} \frac{M^2}{L_{n+1}^2} \quad (k \in D_{n+1}). \quad (3.11)$$

Summing the six coefficients with their common derivative weight L_{n+1} proves (3.4).

Under the common-time hypothesis, these estimates apply simultaneously to all $m \geq 1$. Since $L_m \geq (13/\sqrt{3})2^m$,

$$\sum_{m>N} L_m^{-1} \leq \frac{\sqrt{3}}{13} \sum_{m=N+1}^{\infty} 2^{-m} = \frac{\sqrt{3}}{13} 2^{-N}.$$

Multiplication by $5\sqrt{14}M^2/2$ proves (3.5). The same summable majorant controls the field itself because $L_m \geq 1$. The Weierstrass criterion therefore gives uniform absolute convergence of the Fourier series and its first derivatives. Disjoint supports identify the sum as the original solution's projection on $\bigcup_m D_m$. The bottom layer satisfies

$$|h_0(t)|_{\mathcal{A}_1} \leq 6\sqrt{61}M/\sqrt{6} = \sqrt{366}M.$$

Adding it to (3.5) with $N = 0$ proves (3.6). □

Corollary 3.2 (Different reception times). *For one fixed solution, suppose layer $m \geq 1$ satisfies the three conditions at some existing time t_m , with a legal starting time $s_{m-1} < t_m$. Then*

$$\|\nabla h_m(t_m)\|_\infty \leq \frac{5\sqrt{14}}{2} \frac{M^2}{L_m} \longrightarrow 0.$$

This conclusion does not require the t_m to coincide.

Proof. Apply (3.4) separately at each (s_{m-1}, t_m) , using the same M and the explicit growth of L_m . $\square$

This corollary controls the newly received layer at its own reception time. It does not permit the fields $h_m(t_m)$ to be added and interpreted as a single-time state. If the common-time assumptions of Theorem 3.1 hold at every point of an already existing time set, then (3.5)–(3.6) are uniform on that set. Neither statement controls the complementary projection of u or extends the classical existence interval.

4 The cross-edge chain with two source conditions

This is a second prescribed path, not the next stage of the previous one. Define

$$p_j^{(0)} = R^j(3, 4, 6), \quad p_j^{(n+1)} = 2p_j^{(n)} + p_{j+1}^{(n)} - p_{j+2}^{(n)}. \quad (4.1)$$

In this section only, use

$$\begin{aligned} D_n &= \{\pm p_j^{(n)} : 0 \leq j < 3\}, & h_n &= P_{D_n} u, \\ E_n &= \{\pm(p_i^{(n)} + p_j^{(n)}), \pm(p_i^{(n)} - p_j^{(n)}) : 0 \leq i < j < 3\}, & b_n &= P_{E_n} u. \end{aligned} \quad (4.2)$$

There are three sum pairs and three difference pairs in E_n . For $0 \leq s < a < t < T$, put

$$\begin{aligned} g_{n,s}(r) &= \int_s^r e^{(r-q)\Delta} P_{E_n} B(h_n(q), h_n(q)) \, dq, & s &\leq r \leq t, \\ e_{n,s}(r) &= b_n(r) - g_{n,s}(r), \end{aligned} \quad (4.3)$$

and

$$\begin{aligned} S_{n+1}(t; a) &= \int_a^t e^{(t-r)\Delta} P_{D_{n+1}} B(b_n(r), b_n(r)) \, dr, \\ Z_{n+1}(t; a) &= h_{n+1}(t) - S_{n+1}(t; a). \end{aligned} \quad (4.4)$$

Definition 4.1 (Two-stage source conditions). The first condition requires, for every pair $F \subset E_n$ and every $r \in [a, t]$,

$$\|P_F e_{n,s}(r)\|_{\mathcal{A}_0} \leq \frac{1}{4} \|P_F g_{n,s}(r)\|_{\mathcal{A}_0}. \quad (4.5)$$

The second condition requires, for every pair $F \subset D_{n+1}$,

$$\|P_F Z_{n+1}(t; a)\|_{\mathcal{A}_0} \leq \frac{1}{4} \|P_F S_{n+1}(t; a)\|_{\mathcal{A}_0}. \quad (4.6)$$

All pairs in this layer share the same s, a, t . In particular, the first condition holds throughout the interval where the second source uses the actual field b_n .

Theorem 4.2 (Cross-edge decay). *Under the assumptions of Section 2, if Definition 4.1 holds at layer n on an existing interval $0 \leq s < a < t < T$, then*

$$\|\nabla h_{n+1}(t)\|_\infty \leq |h_{n+1}(t)|_{\dot{A}_1} \leq \frac{125}{672} M^4 4^{-n}. \quad (4.7)$$

If one common existing time t admits, for every $n \geq 0$, times $0 \leq s_n(t) < a_n(t) < t$ satisfying these two-stage conditions, then

$$\sum_{m \geq N} |h_m(t)|_{\dot{A}_1} \leq \frac{125}{504} M^4 4^{-N}, \quad N \geq 0, \quad (4.8)$$

and the chain projection $v_{\text{cross}}(t) = \sum_{m \geq 0} h_m(t)$ belongs to $C^1(\Omega)$, with

$$\|\nabla v_{\text{cross}}(t)\|_\infty \leq \sqrt{366} M + \frac{125}{504} M^4. \quad (4.9)$$

4.1 All-layer geometry

Use A, w_j from (3.7) and write $\mathcal{T} = 2I + R - R^2$. On the axial line $\mathbb{R}(1, 1, 1)$, $\mathcal{T}$ is multiplication by two. On its orthogonal plane, $R^2 + R + I = 0$. Thus $J = R - R^2$ satisfies $J^* = -J$, $J^2 = -3I$, and $\mathcal{T}^* \mathcal{T} = (2I - J)(2I + J) = 7I$ there. Induction gives

$$p_j^{(n)} = 2^n A + \mathcal{T}^n w_j, \quad K_n^2 := |p_j^{(n)}|^2 = \frac{169 \cdot 4^n + 14 \cdot 7^n}{3}, \quad |p_i^{(n)} - p_j^{(n)}|^2 = 14 \cdot 7^n. \quad (4.10)$$

The transverse dot products are those in (3.7) multiplied by 7^n . Hence the sum and difference lengths are

$$\begin{aligned} A_n^+ &:= |p_i^{(n)} + p_j^{(n)}|^2 = \frac{676 \cdot 4^n + 14 \cdot 7^n}{3} \geq K_n^2, \\ A_n^- &:= |p_i^{(n)} - p_j^{(n)}|^2 = 14 \cdot 7^n \geq \frac{14}{61} K_n^2, \quad K_n^2 \geq 61 \cdot 4^n. \end{aligned} \quad (4.11)$$

For the second inequality, multiplication by $3 \cdot 61$ reduces the claim to $169 \cdot 7^n \geq 169 \cdot 4^n$; the last follows in the same way. The matrix $\mathcal{T}$ is invertible on the axis and the transverse plane. The three vectors are consequently independent at every layer, by the same argument as in Section 3. Integer recurrence gives integer frequencies; the coordinate sums $\pm 13 \cdot 2^n$ show that the six-mode D_n are pairwise disjoint. Independence also shows that all twelve frequencies in E_n are distinct.

4.2 The two heat integrations

Fix a layer and suppress its superscript on the p_j . A target in E_n has exactly the two ordered inputs corresponding to its indicated edge, with conjugate coefficients where a negative input occurs. This follows by expressing the target in the independent basis (p_0, p_1, p_2) . Each input frequency has length K_n , and each coefficient has length at most $M/\sqrt{6}$ by (2.9). Therefore

$$|B(h_n, h_n)_k(r)| \leq \frac{K_n M^2}{3}, \quad k \in E_n. \quad (4.12)$$

Both ordered Leray terms are included. The first real-time integral gives

$$|g_{n,s}(r)_k| \leq \frac{K_n M^2}{3|k|^2} (1 - e^{-|k|^2(r-s)}). \quad (4.13)$$

Using (4.11) and then the first source condition on $[a, t]$ yields the following bounds for actual intermediate coefficients:

$$|b_n(r)_{p_i+p_j}| \leq \frac{5M^2}{12K_n}, \quad |b_n(r)_{p_i-p_j}| \leq \frac{305M^2}{168K_n}, \quad a \leq r \leq t. \quad (4.14)$$

Indeed, the ungated source bounds are $M^2/(3K_n)$ and $61M^2/(42K_n)$, respectively, and the condition multiplies each by $5/4$.

Let (i, j, k) be cyclically ordered. The next target is $m = 2p_i + p_j - p_k$. Every element of E_n has exactly two nonzero coordinates, each equal to 1 or -1 , in the independent p basis. For two such elements to sum to $(2, 1, -1)$ in the order (i, j, k) , both must have i coordinate 1, and their remaining coordinates must be $+j$ and $-k$. Thus the complete list of ordered inputs is

$$(p_i + p_j, p_i - p_k), \quad (p_i - p_k, p_i + p_j). \quad (4.15)$$

With $q_+ = p_i + p_j$, $q_- = p_i - p_k$, $U = b_n(r)_{q_+}$ and $V = b_n(r)_{q_-}$, the full coefficient is

$$B(b_n, b_n)_m = -i\mathbb{P}_m[(U \cdot q_-)V + (V \cdot q_+)U].$$

Since $U \cdot q_+ = V \cdot q_- = 0$, the dot products can be written as $U \cdot m$ and $V \cdot m$. By (4.14),

$$|B(b_n, b_n)_m(r)| \leq 2K_{n+1}|U||V| \leq \frac{1525}{1008} \frac{K_{n+1}M^4}{K_n^2}. \quad (4.16)$$

The factor is $2(5/12)(305/168)$. A second integration with the exact target heat multiplier gives

$$|S_{n+1}(t; a)_m| \leq \frac{1525}{1008} \frac{M^4}{K_n^2 K_{n+1}} (1 - e^{-K_{n+1}^2(t-a)}). \quad (4.17)$$

The second pairwise condition, followed by summation of the six target modes, proves

$$|u_m(t)| \leq \frac{7625}{4032} \frac{M^4}{K_n^2 K_{n+1}}, \quad |h_{n+1}(t)|_{\mathcal{A}_1} \leq \frac{7625}{672} \frac{M^4}{K_n^2}. \quad (4.18)$$

Together with $K_n^2 \geq 61 \cdot 4^n$ and $7625 = 61 \cdot 125$, this is (4.7).

Under the common-time hypothesis, summing (4.7) for $n \geq N$ gives $125M^4 4^{-N}/504$, because $\sum_{n=N}^{\infty} 4^{-n} = (4/3)4^{-N}$. This proves (4.8). Frequency lengths are at least one, so the field and its first spatial derivatives converge absolutely and uniformly. Adding the same bottom-layer bound $\sqrt{366}M$ proves (4.9), and completes the proof of Theorem 4.2.

4.3 The complete remainders and the time qualification

For clarity, putting $z_n = u - h_n$ and $w_n = u - b_n$ gives both exact remainders explicitly:

$$\begin{aligned} e_{n,s}(r) &= e^{(r-s)\Delta} b_n(s) \\ &\quad + \int_s^r e^{(r-q)\Delta} P_{E_n}[B(h_n, z_n) + B(z_n, h_n) + B(z_n, z_n)](q) dq, \end{aligned} \quad (4.19)$$

$$\begin{aligned} Z_{n+1}(t; a) &= e^{(t-a)\Delta} h_{n+1}(a) \\ &\quad + \int_a^t e^{(t-r)\Delta} P_{D_{n+1}}[B(b_n, w_n) + B(w_n, b_n) + B(w_n, w_n)](r) dr. \end{aligned} \quad (4.20)$$

In particular, an existing intermediate field at s or target at a is not reset to zero. The second source uses $B(b_n, b_n)$, not $B(g_{n,s}, g_{n,s})$. The identity $B(b_n, b_n) - B(g_{n,s}, g_{n,s}) =$

$B(g_{n,s}, e_{n,s}) + B(e_{n,s}, g_{n,s}) + B(e_{n,s}, e_{n,s})$ does not by itself provide a favorable vector alignment or a relative quadratic error estimate.

If each layer satisfies the two conditions only at its own reception time t_{n+1} , (4.7) still gives $\|\nabla h_{n+1}(t_{n+1})\|_\infty \rightarrow 0$. The fields at those different times cannot be summed to obtain (4.9). A first condition known only at the preparation endpoint a does not supply (4.14) on $[a, t]$ and is not sufficient for this proof. These restrictions are part of the theorem's hypotheses, not conclusions about conditions satisfied by an actual flow.

5 A short-time source-lag obstruction

The chain theorems do not assert their source conditions. The following lemma gives a situation where a selected condition must fail, uniformly over all starting times inside one original short interval.

Lemma 5.1 (Source lag). *Let u be a classical solution as in Section 2, and let $h = P_D u$, where D is finite, $K_D > 0$, and $|k| \leq K_D$ for $k \in D$. Let E_0, E_1, E_2 be disjoint frequency pairs. Suppose that positive constants H, Q, d, T_0 and fields F_j supported on E_j satisfy*

$$\|h(r)\|_{\mathcal{A}_0} \leq Hr^3 \quad (0 \leq r \leq T_0), \quad (5.1)$$

$$\|P_{E_j} u(t) - tF_j\|_{\mathcal{A}_0} \leq Qt^2, \quad \|F_j\|_{\mathcal{A}_0} \geq d \quad (0 \leq t \leq T_0). \quad (5.2)$$

Put

$$C = \frac{K_D H^2}{7}, \quad \tau = \min \left\{ T_0, 1, \frac{d}{2Q}, \frac{d}{8C} \right\}. \quad (5.3)$$

For $E = E_j$ or $E = E_0 \cup E_1 \cup E_2$, and every $0 \leq s < t \leq \tau$, define

$$S_{s,E}(t) = \int_s^t e^{(t-r)\Delta} P_E B(h(r), h(r)) dr, \quad Z_{s,E}(t) = P_E u(t) - S_{s,E}(t).$$

Then

$$\|P_E u(t)\|_{\mathcal{A}_0} > 0, \quad \|S_{s,E}(t)\|_{\mathcal{A}_0} \leq \frac{1}{4} \|P_E u(t)\|_{\mathcal{A}_0}, \quad (5.4)$$

$$\|Z_{s,E}(t)\|_{\mathcal{A}_0} \geq \frac{3}{4} \|P_E u(t)\|_{\mathcal{A}_0} \geq 3 \|S_{s,E}(t)\|_{\mathcal{A}_0}. \quad (5.5)$$

Thus the source condition of Definition 2.1 fails on each pair and on their union. When $S_{s,E}(t) = 0$, its remainder is nonzero; no quotient with zero denominator is used.

Proof. The convolution bound and the input support give

$$\|B(h(r), h(r))\|_{\mathcal{A}_0} \leq \|h(r)\|_{\mathcal{A}_0} |h(r)|_{\mathcal{A}_1} \leq K_D H^2 r^6.$$

Fourier projection and the heat semigroup are contractions in $\mathcal{A}_0$. Consequently, for each permitted E ,

$$\|S_{s,E}(t)\|_{\mathcal{A}_0} \leq C(t^7 - s^7) \leq Ct^7. \quad (5.6)$$

On each pair, (5.2) gives $\|P_{E_j} u(t)\|_{\mathcal{A}_0} \geq dt - Qt^2 \geq dt/2$. Since $t \leq 1$ and $Ct \leq d/8$, we have $Ct^7 \leq dt/8$. This proves (5.4) separately on the three pairs. The $\mathcal{A}_0$ norm is additive across disjoint Fourier supports; summing these three inequalities proves the assertion on their union. The reverse triangle inequality proves (5.5). If $z = u - h$, the exact remainder is

$$\begin{aligned} Z_{s,E}(t) &= e^{(t-s)\Delta} P_E u(s) \\ &\quad + \int_s^t e^{(t-r)\Delta} P_E \{B(h, z) + B(z, h) + B(z, z)\}(r) dr. \end{aligned} \quad (5.7)$$

Thus an existing target at time s has not been discarded. $\square$

Remark 5.2. The bound in (5.6) is obtained before any temporal cancellation. It does not use the generally false inequality $\|S_{s,E}(t)\| \leq \|S_{0,E}(t)\|$. The input estimate is in the original time r , not in $r - s$. There is no assertion for arbitrary starting times beyond τ .

5.1 An explicit fixed datum

Let a_* be the twelve-mode real datum specified in Appendix A; this is the same datum as in [3]. In this subsection set

$$\begin{aligned} p &= (2, 3, 5), \quad \kappa = (3, 4, 6), \quad q = p - Rp = (-3, 1, 2), \\ D &= \{\pm R^j \kappa : 0 \leq j \leq 2\}, \quad E_j = \{\pm R^j q\}, \quad E_- = \bigcup_{j=0}^2 E_j. \end{aligned} \quad (5.8)$$

Let $\bar{A}, \bar{b}, T_*, \delta$ be the explicit positive rational constants in (A.6)–(A.12), and put

$$\begin{aligned} d &= 2\delta, \quad Q_0 = 315\bar{A}^2 + 2\bar{A}\bar{b}^2, \quad C_0 = \frac{32}{7}\bar{A}^2\bar{b}^6, \\ \tau_* &= \min \left\{ T_*, 1, \frac{d}{2Q_0}, \frac{d}{8C_0} \right\} > 0. \end{aligned} \quad (5.9)$$

Proposition 5.3 (A difference output precedes receiver self-supply). *The datum a_* has a unique smooth unforced viscosity-one solution on $[0, T_*]$. For this same solution, with $h = P_D u$, the conclusions (5.4)–(5.5) hold for every $0 \leq s < t \leq \tau_*$, separately for each E_j and for E_- . In particular, the selected D -self-interaction cannot satisfy the source-dominance condition for these difference outputs in this interval.*

We supply the local bounds used in this application, so the proposition does not rely on an unreported numerical evolution or a finite truncation.

Lemma 5.4 (Finite-support local series). *Let a be a real, mean-zero, divergence-free trigonometric polynomial, let $A = \|a\|_{\mathcal{A}_0}$, and suppose its Fourier support F satisfies $|k| \leq K$. Define*

$$v_0(t) = e^{t\Delta} a, \quad v_n(t) = \sum_{j=0}^{n-1} \int_0^t e^{(t-r)\Delta} B(v_j(r), v_{n-1-j}(r)) \, dr \quad (n \geq 1). \quad (5.10)$$

If $4AKt < 1$, then $u(t) = \sum_{n \geq 0} v_n(t)$ is the classical local solution, and

$$\text{supp } \hat{v}_n \subset \underbrace{F + \cdots + F}_{n+1 \text{ summands}}, \quad \|v_n(t)\|_{\mathcal{A}_0} \leq A(4AKt)^n. \quad (5.11)$$

In particular, if $A \leq \bar{A}$, $4AK \leq \bar{b}$, and $\bar{b}t \leq 1/2$, then

$$\sum_{n \geq m} \|v_n(t)\|_{\mathcal{A}_0} \leq 2\bar{A}\bar{b}^m t^m \quad (m \geq 1). \quad (5.12)$$

Proof. The support assertion follows by induction from convolution and the heat multiplier. Let $c_0 = 1$ and $c_n = \sum_{j=0}^{n-1} c_j c_{n-1-j}$. These Catalan numbers obey $c_n \leq 4^n$. Inductively, the sharper bound $\|v_n(t)\|_{\mathcal{A}_0} \leq c_n A^{n+1} (Kt)^n$ follows from $|v_{n-1-j}|_{\mathcal{A}_1} \leq (n-j)K \|v_{n-1-j}\|_{\mathcal{A}_0}$ and

$$\frac{1}{n} \sum_{j=0}^{n-1} (n-j) c_j c_{n-1-j} = \frac{n+1}{2n} c_n \leq c_n.$$

For every nonnegative integer m , the support bound also gives

$$\sum_k (1 + |k|)^m |(v_n)_k| \leq [1 + (n+1)K]^m A(4AKt)^n.$$

On a compact time interval with $4AKt < 1$, the right side is summable in n , for all m . Thus the series and all spatial derivatives converge uniformly. The same estimates with two extra powers of the support radius justify differentiating the integral equation once in time; successive differentiation of the equation gives smoothness. Absolute convergence permits the double series for $B(u, u)$ to be regrouped by n , giving the full mild equation. Reality, zero mean and incompressibility are preserved term by term. Recovering the pressure by the complementary Fourier projection gives the unforced Navier–Stokes equation. Classical uniqueness follows by the energy estimate for the difference of two solutions: the transport term of one divergence-free field vanishes, and the remaining term is at most its bounded velocity gradient times the square of the difference’s L^2 norm. Gronwall’s inequality applies. Finally, summing a geometric series proves (5.12). $\square$

Proof of Proposition 5.3. The datum in Appendix A satisfies $A \leq \bar{A}$ and $K = \sqrt{38} < 7$, while $\bar{b} = 28\bar{A}$ and $\bar{b}T_* \leq 1/2$. Lemma 5.4 proves existence and uniqueness throughout the stated closed interval.

For a frequency k , write $\sigma(k) = k_1 + k_2 + k_3$. Every initial frequency has $\sigma = \pm 1$ or ± 10 . A sum of at most three such frequencies cannot have coordinate sum 13: if h and l are the signed numbers of high and low leaves, respectively, then $10h + l = 13$ and $|h| + |l| \leq 3$ have no integer solution. The same argument applies to -13 . Consequently, $P_D v_0 = P_D v_1 = P_D v_2 = 0$ identically in time, and

$$\|h(r)\|_{\mathcal{A}_0} \leq 2\bar{A}\bar{b}^3 r^3 \quad (0 \leq r \leq T_*). \quad (5.13)$$

This is a bound for the actual projection, including every higher-order interaction. For completeness, a four-leaf representation of κ must have one positive high leaf and three positive low leaves. Of the three high choices, only p leaves a nonnegative vector with coordinate sum three, namely $\kappa - p = (1, 1, 1)$.

At $q = p - Rp$, exactly the two ordered initial pairs $(p, -Rp)$ and $(-Rp, p)$ contribute. Indeed $\sigma(q) = 0$ excludes a high–low pair; a difference of low frequencies has all coordinates in $\{-1, 0, 1\}$, and the three high-frequency differences are precisely the three pairs E_j . Write

$$V = (3v_* + 5w_*, -2v_*, -2w_*), \quad N = p \times Rp = (-1, 19, -11).$$

The sum before the Leray projection is

$$X = -(11v_* + 19w_*)RV + (-v_* + 11w_*)V. \quad (5.14)$$

Since $|q|^2 = 14$, a direct projection yields

$$\mathbb{P}_q X = -\frac{4}{7}(3v_* + 2w_*)(v_* + 3w_*)N, \quad B(a_*, a_*)_q = iG, \quad G = \frac{4}{7}(3v_* + 2w_*)(v_* + 3w_*)N. \quad (5.15)$$

One can verify the projection without dropping any component: $q \cdot X = 22v_*^2 - 204v_*w_* - 418w_*^2$ and $\mathbb{P}_q X = X - q(q \cdot X)/14$. Rotation and complex conjugation give the other five coefficients.

Both inputs in this calculation have squared length 38. Thus, if F_j has coefficients $iR^j G$ and $-iR^j G$ at $R^j q$ and $-R^j q$, respectively, then

$$P_{E_j} v_1(t) = \Phi(t) F_j, \quad \Phi(t) = \int_0^t e^{-14(t-r)} e^{-76r} dr = \frac{e^{-14t} - e^{-76t}}{62}. \quad (5.16)$$

For $0 \leq r \leq t$, the inequality $0 \leq 1 - e^{-x} \leq x$ for $x \geq 0$ gives

$$0 \leq t - \Phi(t) \leq \int_0^t [14(t-r) + 76r] dr = 45t^2.$$

Also $P_{E_j}v_0 = 0$ and $\|F_j\|_{\mathcal{A}_0} \leq \|B(a_*, a_*)\|_{\mathcal{A}_0} \leq 7\bar{A}^2$. Hence (5.12) with $m = 2$ gives

$$\|P_{E_j}u(t) - tF_j\|_{\mathcal{A}_0} \leq (315\bar{A}^2 + 2\bar{A}\bar{b}^2)t^2 = Q_0t^2. \quad (5.17)$$

Appendix A verifies $|G_1| \geq \delta > 0$; therefore $\|F_j\|_{\mathcal{A}_0} = 2|G| \geq 2\delta = d$ on every pair. Finally $|k| = \sqrt{61} < 8$ on D . Apply Lemma 5.1 with $K_D = 8$, $H = 2\bar{A}\bar{b}^3$, $Q = Q_0$ and $T_0 = T_*$. Its constant is exactly C_0 , and its interval is (5.9). $\square$

The companion result proves an earlier source-selection property for this same datum and the same receiver D , using its actual first generated band [3]. Proposition 5.3 addresses a different source: the self-interaction of the receiver D into E_- . It does not negate that earlier property, nor assert that a general regeneration mechanism fails at later times.

6 Scope, proof ingredients and computational assistance

Both chain arguments combine estimates on selected Duhamel sources with conditions comparing each source with its complete remainder. The adjacent-sum source and the first cross-edge source are bounded unconditionally. In the cross-edge argument, the first condition on the whole second integration interval is needed to bound the second source. The corresponding target conditions then give the actual target gradient bounds. If a required condition fails, the calculation has not shown that the target is small. The remainder can contain sources outside the chosen sparse chain, including interactions with already generated frequencies.

The assertions keep the domain, viscosity, solution and initial datum fixed. They allow different starting and reception times at different layers, but these times must belong to that solution's classical interval. Summing layer bounds requires the stronger common-time conditions stated in the theorems. Even then the resulting regularity concerns the chain projection. No estimate for its complementary Fourier projection, no continuation criterion for the full velocity and no exclusion of all possible singularity mechanisms is established here.

The first chain uses its exact all-layer frequency differences, not a generic bound by the sending frequency. In the second chain those differences grow, but the first heat integration controls both prescribed edge types and the second controls their subsequent interaction. The first source condition must hold throughout that second integral. An endpoint comparison cannot replace it, and replacing the actual edge field by its selected source would require a further cancellation estimate which is not assumed or proved here.

The short-time example has a separate purpose. Its nonzero difference response is first order in the original time, whereas the selected receiver is bounded by a third-order tail. This mismatch gives a uniform obstruction for all truncated integrals within one explicitly specified short interval. It does not reset the initial datum at a new time or extend that interval by translation.

Exact calculations and reproducibility. The two chain proofs use analytic estimates and finite-dimensional vector identities displayed in the text. The fixed-datum example additionally uses exact rational polynomial arithmetic, Sturm root counts and rational interval evaluation. Appendix A gives the polynomials, intervals, constants and checking

procedures needed for its application. The finite calculations were developed and checked with exact-arithmetic programs; floating-point evolution is not used as proof of any assertion. The analytic estimates control the infinite Fourier series. Raw development logs and general research programs are not public supplementary material for this manuscript.

AI assistance and responsibility. Generative AI tools assisted with drafting, literature retrieval, proof organization, symbolic derivations and verification scripts. Exact finite calculations and automated checks are distinct from the analytic arguments presented here. They are not human peer review or formal proof-assistant certification. The author is responsible for the manuscript and its claims.

A Explicit algebraic datum and rational checks

This appendix supplies the finite data used only in Proposition 5.3. The two chain theorems do not depend on these polynomials or on an algebraic choice of initial datum.

A.1 The datum and the local constants

Define

$$P(X) = \frac{12379608041588}{607290075} + \frac{129169813631527}{1012150125}X + \frac{111813118362046}{337383375}X^2 \\ + \frac{1377133612024132}{3036450375}X^3 + \frac{207263796033514}{607290075}X^4 \\ + \frac{401161881122159}{3036450375}X^5 + \frac{4124532235492}{202430025}X^6, \quad (\text{A.1})$$

$$r(X) = \frac{1807613971}{12393675} + \frac{9523582624}{20656125}X + \frac{27816146846}{61968375}X^2 + \frac{196585982}{1770525}X^3, \quad (\text{A.2})$$

$$s(X) = \frac{536919079}{2699067} + \frac{9469267576}{13495335}X + \frac{10524885968}{13495335}X^2 + \frac{646271818}{2699067}X^3. \quad (\text{A.3})$$

Let α_* be the unique root of P in $I_0 = (-75/32, -125/64]$, and put

$$v_* = -s(\alpha_*), \quad w_* = r(\alpha_*), \quad U = (0, 1, \alpha_*), \quad V = (3v_* + 5w_*, -2v_*, -2w_*). \quad (\text{A.4})$$

The root assertion is checked below; irreducibility of P is not needed. With $R(x_1, x_2, x_3) = (x_3, x_1, x_2)$ and $p = (2, 3, 5)$, define

$$a_*(x) = \sum_{j=0}^2 [R^j U \{e^{i(R^j e_1) \cdot x} + e^{-i(R^j e_1) \cdot x}\} \\ + R^j V \{e^{i(R^j p) \cdot x} + e^{-i(R^j p) \cdot x}\}]. \quad (\text{A.5})$$

No amplitude is subsequently rescaled. These are twelve fixed Fourier coefficients. Reality and zero mean are immediate, and $e_1 \cdot U = p \cdot V = 0$ proves incompressibility on every rotated frequency. The datum obeys $a_*(Rx) = Ra_*(x)$.

Use the positive rational constants

$$\bar{A} = \frac{13990473994187561}{829153382400}, \quad \bar{b} = 28\bar{A}, \quad A_\alpha = \frac{75}{32}, \\ \ell = \frac{32000398665717304027531027}{477317255069800857600}, \\ \bar{E} = 80\bar{A}^4\bar{b}^5 + 3\bar{A}\bar{b}^4, \quad D_f = 2(1 + A_\alpha^2) + 2\bar{A}\bar{b}^2, \\ T_* = \min \left\{ \frac{1}{2\bar{b}}, \frac{\ell}{8\bar{E}}, \frac{1}{2D_f} \right\}. \quad (\text{A.6})$$

This retains the same conservative interval used for the datum in [3]. In the present paper only $T_* > 0$ and $\bar{b}T_* \leq 1/2$ are needed; both follow directly from this definition.

On the closure of I_0 , rational interval evaluation gives

$$\begin{aligned} |s(X)| \leq U_v &= \frac{39312542562313}{88443027456}, & |r(X)| \leq U_w &= \frac{19711525523669}{58016563200}, \\ |-3s(X) + 5r(X)| \leq U_1 &= \frac{585728660549}{472227840}. \end{aligned} \quad (\text{A.7})$$

These constants satisfy the exact identity

$$\bar{A} = 6(1 + A_\alpha + U_1 + 2U_v + 2U_w).$$

The Euclidean norm of a vector is at most the sum of its coordinate absolute values. Therefore

$$\|a_*\|_{\mathcal{A}_0} = 6(|U| + |V|) \leq 6(1 + |\alpha_*| + |-3s(\alpha_*) + 5r(\alpha_*)| + 2|s(\alpha_*)| + 2|r(\alpha_*)|) \leq \bar{A}.$$

All initial frequencies have length at most $\sqrt{38} < 7$.

A.2 A nonzero coordinate of the difference response

The first coordinate of G in (5.15) is

$$f(\alpha_*), \quad f(X) = -\frac{4}{7}(-3s(X) + 2r(X))(-s(X) + 3r(X)). \quad (\text{A.8})$$

For an exact replay of the interval calculation, write $f(X) = \sum_{j=0}^6 f_j X^j$. Its rational coefficients are

$$\begin{aligned} f_0 &= \frac{35798055470904413384872}{860536215451513125}, & f_1 &= \frac{401729094518327896325536}{1434227025752521875}, \\ f_2 &= \frac{16265521854261904842210248}{21513405386287828125}, & f_3 &= \frac{22128606310912506680936608}{21513405386287828125}, \\ f_4 &= \frac{15565564652878394325142432}{21513405386287828125}, & f_5 &= \frac{1023532801151109314237936}{4302681077257565625}, \\ f_6 &= \frac{22854636325453540780192}{860536215451513125}. \end{aligned} \quad (\text{A.9})$$

The same root lies in the refined interval

$$I_5 = (-2075/1024, -4125/2048]. \quad (\text{A.10})$$

Horner interval evaluation of (A.9) on the closure of I_5 gives

$$f(\alpha_*) \in [-N_-/D_f, -N_+/D_f], \quad (\text{A.11})$$

where the integers are

$$\begin{aligned} N_- &= 29437612184526502853366735142169004680277, \\ N_+ &= 3880728698927672352860026215467141443157, \\ D_f &= 248032677071760087808993803632640000. \end{aligned}$$

In particular, the explicitly positive rational number

$$\delta = \frac{N_+}{D_f} \quad (\text{A.12})$$

satisfies $|G_1| = |f(\alpha_*)| \geq \delta > 0$. This is a lower bound from one coordinate, not an evaluation of the full Hermitian vector norm. The rotation and conjugation identities in (5.15) give the same pair bound $d = 2\delta$ for all three pairs.

A.3 Exact verification procedure

For clarity, the rational checks behind the preceding statements can be performed using only the displayed coefficients. Polynomial addition, multiplication and long division are taken in $\mathbb{Q}[X]$, with trailing zero coefficients removed. To form a Sturm sequence, start with P, P' and successively append the negative remainder of the preceding two polynomials, stopping at zero. An entry may be multiplied by a positive rational number without changing the sign variations. At a rational endpoint evaluate every entry by Horner's rule, delete zero values and count changes of sign. At the two endpoints of I_0 the variation numbers are 4 and 3, respectively; neither endpoint is a root of P . Sturm's theorem therefore gives the required unique root in I_0 . The root counts in its five successive bisections are displayed in Table 1. None of the listed midpoints is a root. This proves (A.10) for the original fixed root.

Step	Midpoint	Left count	Right count	Retained interval
1	$-275/128$	0	1	$(-275/128, -125/64]$
2	$-525/256$	0	1	$(-525/256, -125/64]$
3	$-1025/512$	1	0	$(-525/256, -1025/512]$
4	$-2075/1024$	0	1	$(-2075/1024, -1025/512]$
5	$-4125/2048$	1	0	$(-2075/1024, -4125/2048]$

Table 1: Exact bisections retaining the same root. Counts refer to the left and right subintervals of the retained interval from the preceding step.

For a closed rational interval $I = [a, b]$, addition is endpointwise and multiplication of $[a, b]$ by $[c, d]$ is the interval with endpoints the minimum and maximum of ac, ad, bc, bd . Evaluate a polynomial $\sum_{j=0}^m c_j X^j$ by starting with $[c_m, c_m]$ and replacing the current interval J by $JI + [c_j, c_j]$, for $j = m - 1, \dots, 0$. Every step encloses all exact polynomial values on I . Apply this rule to s, r and the expanded polynomial $-3s + 5r$ on the closure of I_0 and take the larger absolute endpoint; this gives (A.7). Expanding the two products in (A.8) gives exactly (A.9); applying the same interval rule to those coefficients on the closure of I_5 gives (A.11).

These procedures involve finite rational operations and the elementary Sturm root-count theorem. They do not require a minimal polynomial for α_* , a floating-point approximation to that root, or any numerical solution of Navier–Stokes. They establish the lower bound and the positive window used in Proposition 5.3; no later-time extension follows from the finite calculations.

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